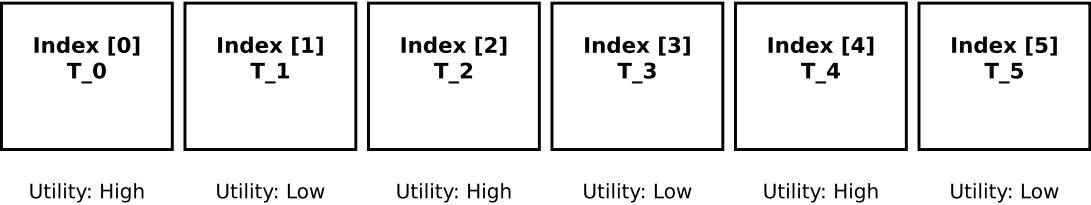


STATE A: BEFORE COMPACTION (FRAGMENTED BUFFER)



ASSOCIATIVE TARGET MAP f(x) & DISJOINT CYCLE RESOLUTION

f(0) = 0 [Fixed Point / Stabilized Attractor Basin - No Movement]

f(1)=3, f(2)=1, f(3)=5, f(4)=2, f(5)=4

Identified Disjoint Cycle: (1 -> 3 -> 5 -> 4 -> 2 -> 1)

IN-PLACE CYCLE EXECUTION STEPS:

- 1. Temp_Reg <-- Load(Block 1)
- 2. Block 1 <-- Block 2 (f(2)=1)
- 3. Block 2 <-- Block 4 (f(4)=2)
- 4. Block 4 <-- Block 5 (f(5)=4)
- 5. Block 5 <-- Block 3 (f(3)=5)
- 6. Block 3 <-- Temp_Reg (Cycle Closed)

Temp_Reg [O(1) Memory]

Holds Payload of Index 1

STATE B: AFTER COMPACTION (CONTIGUOUS ACTIVE BUFFER)

Allocation Boundary Pointer
(Buffer Truncated Here)

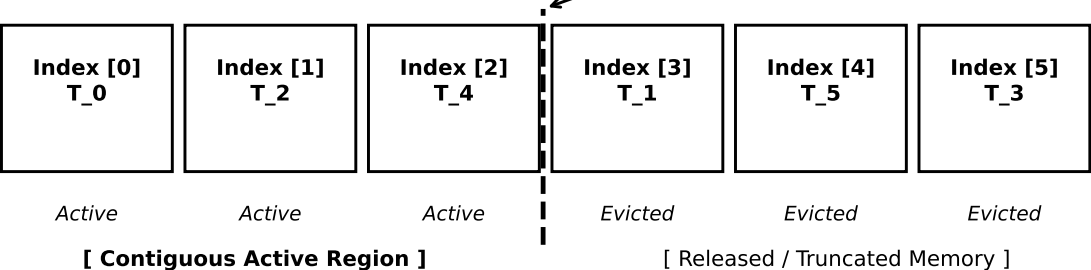


FIG. 2